PayGuard AI

Complete Implementation Guide

Pre-Authorization Risk Intelligence Platform for PayPal Merchants

Project Type:	PayPal AI Hackathon 2026
Version:	1.0
Date:	November 3, 2026
Status:	Implementation Ready

Executive Summary

PayGuard AI is a pre-transaction risk screening platform that protects merchants from PayPal account freezes by predicting and filtering high-risk transactions before they reach PayPal's systems. Unlike PayPal's existing fraud protection (which is reactive), PayGuard AI operates proactively to maintain merchant account health.

Key Differentiation:

- PayPal Fraud Protection: Post-transaction fraud detection and reimbursement
- PayGuard AI: Pre-transaction risk screening and intelligent routing

Value Proposition:

- Prevents account freezes that cost merchants an average of $47,000 per incident
- Reduces chargeback ratios by filtering risky transactions upstream
- Maintains healthy PayPal account metrics for sustainable growth
- Routes 90% of clean transactions through PayPal, 10% high-risk to alternative gateways

Table of Contents

1. Architecture Overview
2. Technology Stack
3. PayPal API Integration
4. AI/ML Risk Engine
5. Implementation Roadmap
6. False Positive Analysis
7. Challenges & Mitigations
8. Demo Strategy
9. Business Model

1. Architecture Overview

System Flow

PayGuard AI intercepts transactions at the checkout stage, before they reach PayPal's systems. The platform analyzes 50+ risk signals in real-time (<100ms) to categorize transactions into three risk tiers: GREEN (low risk - proceed normally), YELLOW (medium risk - require additional verification), and RED (high risk - block or route to alternative gateway).

Core Components

Component	Purpose	Technology
Risk Analysis Engine	Real-time risk scoring using ML and pattern matching	XGBoost, TensorFlow.js, Elastic
Feature Extraction	Extract 50+ risk signals from transaction data	Node.js, Redis
PayPal Integration	Connect with PayPal APIs for checkout and webhooks	PayPal SDK, REST APIs
Merchant Dashboard	Real-time monitoring and account health tracking	Next.js, AG Grid, Bryntum
Feedback Loop	Continuous learning from transaction outcomes	Webhooks, PostgreSQL

Risk Categorization

Risk Level	Score Range	Action	Expected %
GREEN	0.0 - 0.3	Direct to PayPal - instant checkout	90%
YELLOW	0.3 - 0.7	Request additional verification (2FA, SMS)	7%
RED	0.7 - 1.0	Block or route to alternative gateway	3%